

Ionic Liquid Selectively Facilitates CO₂ Transport through Graphene Oxide Membrane

Wen Ying,[†] Jingsong Cai,[‡] Ke Zhou,[§] Danke Chen,[†] Yulong Ying,[†] Yi Guo,[†] Xueqian Kong,^{*,‡} Zhiping Xu,^{*,§} and Xinsheng Peng^{*,†}

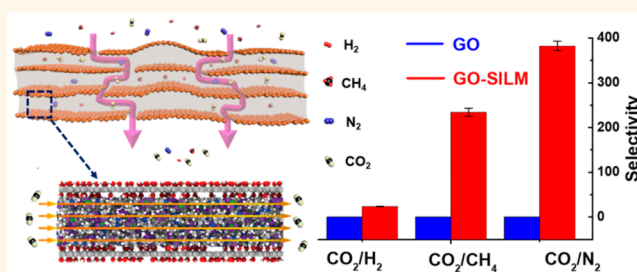
[†]State Key Laboratory of Silicon Materials, Department of Materials Science and Engineering, and [‡]Department of Chemistry, Zhejiang University, Hangzhou 310027, China

[§]Applied Mechanics Laboratory, Department of Engineering Mechanics and Center for Nano and Micro Mechanics, Tsinghua University, Beijing 100084, China

S Supporting Information

ABSTRACT: Membrane separation of CO₂ from H₂, N₂, or CH₄ has economic benefits. However, the trade-off between selectivity and permanence in membrane separation is challenging. Here, we prepared a high-performance CO₂-philic membrane by confining the [BMIM][BF₄] ionic liquid to the nanochannels in a laminated graphene oxide membrane. Nanoconfinement causes the [BMIM][BF₄] cations and anions to stratify. The layered anions facilitate CO₂ transportation with a permeance of 68.5 GPU. The CO₂/H₂, CO₂/CH₄, and CO₂/N₂ selectivities are 24, 234, and 382, respectively, which are up to 7 times higher than that of GO-based membranes and superior to the 2008 Robeson upper bound. Additionally, the resultant membrane has a high-temperature resistance, long-term durability, and high-pressure stability, indicating its great potential for CO₂ separation applications. Nanoconfining an ionic liquid into the two-dimensional nanochannels of a laminated membrane is a promising gas separation method and a nice system for investigating ionic liquid behavior in nanoconfined environments.

KEYWORDS: nanoconfinement, graphene oxide-supported ionic liquid membranes, gas separation, two-dimensional nanochannel, solution-diffusion mechanism



Gas separation and enrichment play an important role in modern industries. Separating CO₂ from other light gases (such as H₂, N₂, and CH₄)¹ is a critical process for syngas, natural gas, biogas, and CO₂ capture industries, and lowering the costs of CO₂ separation or capture systems is a challenge. Membrane separation techniques are highly promising due to their low-energy consumption and high efficiency.² However, the performance of current separation membranes still needs to improve for industrial development.³ To overcome the existing weaknesses of membranes and to enhance their performances, various new membrane materials are being developed, such as metal–organic frameworks (MOFs),^{4,5} graphene-based materials,^{6,7} and ionic liquid (IL)-based membranes.^{8,9,16} In addition, using mixed matrix membranes (MMMs) consisting of polymer and inorganic fillers is also a potential method; for example, oriented graphene oxide (GO) platelets were used to extend the diffusion path to improve the separation performance.¹⁰ An ultrathin GO membrane has been reported to have a high H₂ permeance and selectivity for H₂/CO₂ and H₂/N₂, which is attributed to the nanodefects on GO nanosheets.¹¹ However, a GO membrane with a certain water content shows good CO₂/

H₂, CO₂/CH₄, and CO₂/N₂ selectivity^{12,13} because water can increase the CO₂ permeance and can decrease the permeance of H₂, CH₄, and N₂. Nevertheless, due to the volatility and ultrafast permeation of H₂O through GO membranes¹³ during separation processes, water-wetted GO membranes cannot withstand long-term operations or high temperatures without a continuous water supply.

In contrast, ILs have great potential for gas separation applications because of their extremely low volatilities, excellent chemical and thermal stabilities, and high viscosities.¹⁴ Typically, ILs have higher solubilities for CO₂ than H₂, CH₄, and N₂. However, due to their high cost and liquid state, the direct use of ILs as a membrane for CO₂ separation under pressure is difficult. Therefore, ILs must be supported by a porous, solid substrate with mechanical stability.^{15–18} Supported ionic liquid phase (SILP), which covers the pore surface of a porous medium by a thin layer IL, and supported ionic liquid membrane (SILM), which confines IL into the porous

Received: January 15, 2018

Accepted: June 6, 2018

Published: June 6, 2018

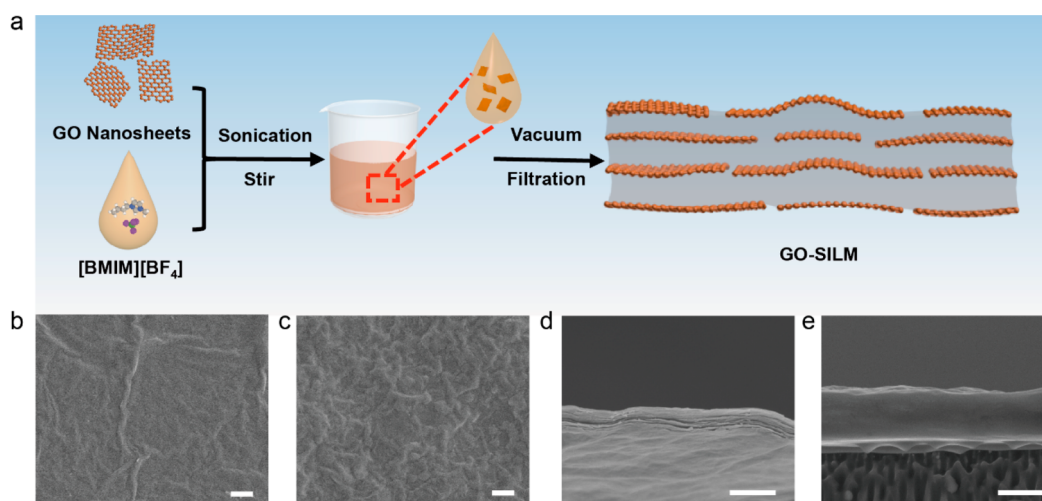


Figure 1. Preparation scheme for GO-SILMs and scanning electron microscopy images of a GO membrane fabricated with 0.75 mL of a 0.2 mg/mL GO aqueous dispersion and a GO-SILM fabricated with 3 mL of a 0.05 mg/mL GO/[BMIM][BF₄] dispersion: (a) the scheme; (b,c) the surface of the GO membrane and GO-SILM; (d,e) the cross section of the GO membrane and GO-SILM, respectively. The scale bars in b–e represent 1 μm .

substrate, are being studied for CO₂ separation and capture.^{14,19} Though significant improvements have been achieved, a trade-off between the selectivity and permeance of SIL membranes with conventional porous supports still exists. Due to the unavoidable leakage of ILs from the large pores in membranes under a relatively high pressure, a thicker support (typically 50–150 μm)⁸ is needed, which obviously limits the permeance. Confining ILs into nanopores could dramatically increase their high-pressure stability and corresponding properties.²⁰ For example, ILs nanoconfined in silica gel,²¹ MOFs,^{15,16} and zeolites^{17,18} have shown enhanced CO₂ separation and capture performances, whereas their fragile, complex formation process may limit their implementation in gas separation membranes. Due to their flexible, mechanically strong, easily scalable properties,^{6,7} GO-laminated membranes can provide a two-dimensional (2D) nanospace to support ILs and enhance their separation performance. Confining ILs into laminated GO membranes, forming GO-supported ionic liquid membrane (GO-SILMs), should overcome the drawbacks, such as the volatility of water or of wetted GO membranes,^{12,13} and improve the separation performance by sealing the defects and large wrinkles¹² in the GO membrane.

In this work, we immobilized 1-butyl-3-methylimidazolium tetrafluoroborate [BMIM][BF₄], an inexpensive room temperature ionic liquid with a high CO₂ solubility and low H₂, CH₄, and N₂ solubilities,²² into the 2D nanochannels of laminated GO membranes and fabricated GO-SILMs to separate CO₂ from other light gases (see scheme in Figure 1a). Due to the high affinity of [BMIM][BF₄], it can be easily introduced and confined into the 2D nanochannels of laminated GO membranes, which was proven by the solid-state NMR (SSNMR) characterization. The resultant GO-SILM has a fast CO₂ permeance and extremely high CO₂/H₂, CO₂/CH₄, and CO₂/N₂ selectivities, which are up to 7 times greater than those of GO-based membranes^{6,8,9} and superior to that of most state-of-the-art membranes.^{23–34} In addition, the resultant GO-SILM also exhibits excellent stability due to the high viscosity and thermal stability of the IL.

RESULTS AND DISCUSSION

Synthesis and Characterization of the GO-SILMs. GO nanosheets were prepared by a modified Hummer's method¹² (see the details in the Methods section). The as-prepared GO nanosheets with a height of 0.75 nm are shown in Figure S1. Figure 1b,d clearly shows the surface wrinkles and layered structure of the GO membrane, which was assembled by vacuum filtration of 0.75 mL of a GO aqueous dispersion (0.2 mg/mL). The thickness of the resultant GO membrane is 470 nm. The contact angle of [BMIM][BF₄] on the GO membrane is approximately 30° (Figure S2). Due to their wettability, the GO nanosheets can be well dispersed in a [BMIM][BF₄] solution by ultrasonication (see the details in the Methods section). The GO-SILM was prepared by vacuum filtration of 3 mL of a GO/[BMIM][BF₄] dispersion (0.05 mg/mL) on an anodic alumina oxide (AAO) membrane. The mass ratio of GO and [BMIM][BF₄] in the GO-SILM membrane is 0.46 (see the detailed calculation process in the Calculation Section in Methods). Compared with those on the GO membrane (Figure 1b), the wrinkles on the GO-SILM surface are much shorter and denser (Figure 1c), resulting in a rougher surface. Moreover, the clear laminated structure in the GO membrane (Figure 1d) cannot be observed in the GO-SILM (Figure 1e) because the layered GO nanosheets are fulfilled by the impregnated IL. This laminated structure can be observed again after removing IL by being treated at 873 K in N₂ gas for 2 h (Figure S3). The thickness of the GO-SILM containing 0.15 mg of GO is 1050 nm, which is much thicker than the GO membrane (470 nm) prepared from 0.15 mg of GO. These results indicate that [BMIM][BF₄] fully filled the GO channels, resulting in a significant interlayer distance shift from 0.87 nm (the pristine GO membrane, calculated from the X-ray diffraction (XRD) pattern, Figure S4) to 1.94 nm (GO-SILM membrane; see the calculation details in the Calculation Section in Methods). The uniform distribution of N, B, and F elements along the cross section of GO-SILM further indicates that [BMIM][BF₄] is fully immobilized in the GO channels (Figure S5).

Fourier transform infrared (FTIR) spectra, differential scanning calorimetry (DSC) curves (Figure S6), and

SSNMR³⁵ spectra (Figure 2) of the [BMIM][BF₄] GO membrane and GO-SILM were recorded. Supporting Informa-

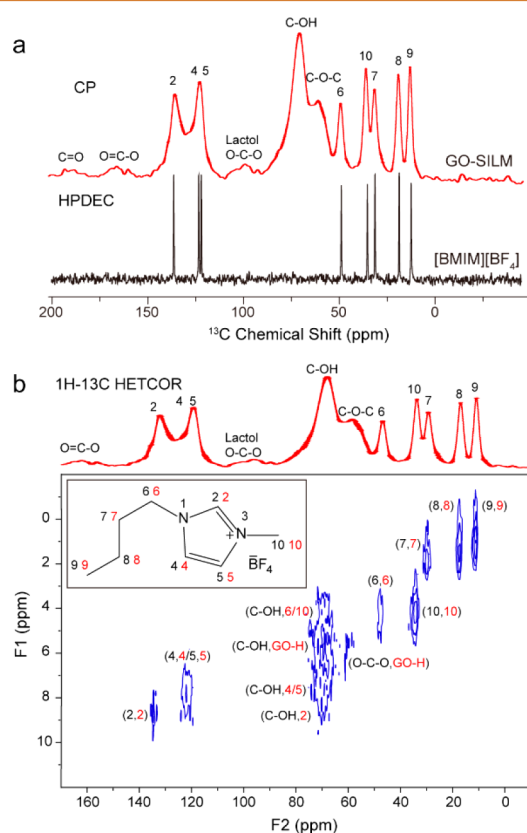


Figure 2. NMR characterization. (a) ¹³C spectra of GO-SILM and [BMIM][BF₄]; (b) ¹H-¹³C spectra of GO-SILM. The inset in b is the molecular structure of a BMIM⁺ ion.

tion Figure S6 shows that most of the characteristic FTIR peaks of [BMIM][BF₄] are observed in the GO-SILM sample,³⁶ which indicates that [BMIM][BF₄] is incorporated into the GO. The peaks associated with the cation red-shifted; the peaks indexed to the stretch of the imidazole ring at 3163.21 and 3122.71 cm⁻¹ had the most obvious red shifts due to the strong confinement. In contrast, the peaks associated with the anion slightly blue-shifted. The GO nanosheets are thought to interact with both the cation and anion, but the interaction with the cation is stronger than that with the anion. The NMR results (Figure 2) further confirm the confinement of the imidazole ring (Figure 2a) and the interaction between the imidazole ring and the oxygen-containing functional groups of GO, especially the hydroxy groups. Figure 2b shows that the GO hydroxy groups obviously correlate with the H2, H4, H5, H6, and H10 of the imidazole ring. Both the electrostatic and π - π interactions between the negatively charged GO and positively charged [BMIM]⁺ group also contribute to the strong association of GO and [BMIM][BF₄].³⁷ The specific FTIR and NMR data are shown in Tables S1 and S2. The nanoconfinement significantly influences the thermodynamic properties. The freezing point is suppressed from -75.4 °C for bulk [BMIM][BF₄] to -69.1 °C for the nanoconfined [BMIM][BF₄] (Figure S6). Based on the Young-Laplace equation,³⁸ the capillary force of filled [BMIM][BF₄] in GO nanochannels (1.94 nm) is calculated to be 82.5 MPa (see the details in the Calculation Section of Methods), which shows a

strong driving force to maintain [BMIM][BF₄] into the channels of GO. This is also confirmed by the high interaction energy by simulation (see the details in the Methods section).

Gas Separation. As mentioned above, nanoconfining ILs can significantly affect the properties of the IL, which may lead to highly efficient gas separation performances. Here, the [BMIM][BF₄] nanoconfined in the 2D nanospace of a laminated GO membrane is used to separate CO₂ from H₂, CH₄, and N₂. Before the gas separation experiment, the as-prepared GO-SILMs on AAO supports were floated on ethanol for 10 s and quickly dried by blowing hot air to remove the IL trapped in the AAO channels, followed by vacuum drying at 60 °C for 12 h with very slightly reduced the GO (see the details in the Methods section and Tables S3 and S4). The gas permeance through the GO-SILMs was measured by a bubble flow meter (Figure S7) under 0.1 MPa (Figure 3) and time-lag method.³⁹ As Figure 3b shows, the H₂/CO₂, CH₄/CO₂, and N₂/CO₂ selectivities of a pristine GO membrane with a thickness of 470 nm are 4, 1.8, and 1.3, respectively, which are very close to the Knudsen selectivity⁴⁰ due to the presence of large, wrinkled nanochannels (Figure 1b). The permeance of CO₂ through GO-SILM is 68.5 gas permeation units (GPU), which is higher than 2.88, 0.293, and 0.179 GPU for H₂, CH₄, and N₂, respectively. The as-calculated selectivities of CO₂/H₂, CO₂/CH₄, and CO₂/N₂ are 24, 234, and 382, respectively, which are the best selectivities among [BMIM][BF₄]-based and GO-based membranes for separating CO₂ from CH₄ and N₂ (Figure 3c,d and Table S5). Upon manipulating the thickness and obtaining the permeability in Barrers, the performance is obviously above the “2008 Robeson upper bound” (Figure S8 and Table S6). The Robeson upper bound mainly focuses on the permeability of a material rather than the real membrane performance. For industrial applications and economic evaluations, the permeance (the pressure normalized flux through a membrane) is more reliable than the permeability. Therefore, a thin membrane with both a high selectivity and permeance is desirable. Clearly, the performances of our GO-SILMs are competitive, and they are better than most of the previously reported membranes for separating CO₂ from other light gases (Figure S9 and Table S6). In contrast to the GO-based membranes,^{6,12,13} impregnated ILs seal the large defects and wrinkles in the GO membrane and significantly improve the CO₂ selectivity based on the distinct solubility difference of ILs for gases. The superior performance of the GO-SILMs compared to that of conventional SILMs is due to the nanoconfinement of the ILs in the 2D channels of the GO membrane with a thickness of only approximately 1 μ m. The molecular dynamics (MD) simulation results confirm the contribution of nanoconfined ILs in the GO-SILM as discussed (see the details in the MD simulation details section in Methods). The results show that the binding energies between gas species (CO₂, N₂, H₂, CH₄) and the confined [BMIM][BF₄] are slightly higher than those for bulk [BMIM][BF₄], which means that gas dissolution in the nanoconfined [BMIM][BF₄] is energetically preferable (Figure S10 and Table S7). Moreover, the binding energy of CO₂ is significantly higher than the others, and the nanoconfinement reduces the self-diffusivities of the gas molecules, with the reduction for CO₂ (57%) being less significant compared to others (Table S7). The increase of solubility and decrease of self-diffusivity are confirmed by sorption and adsorption isotherms (Figure S11) and time-lag method³⁹ (Table S8), respectively. Typically, the CO₂/gas selectivity is mainly dominated by the solubility of

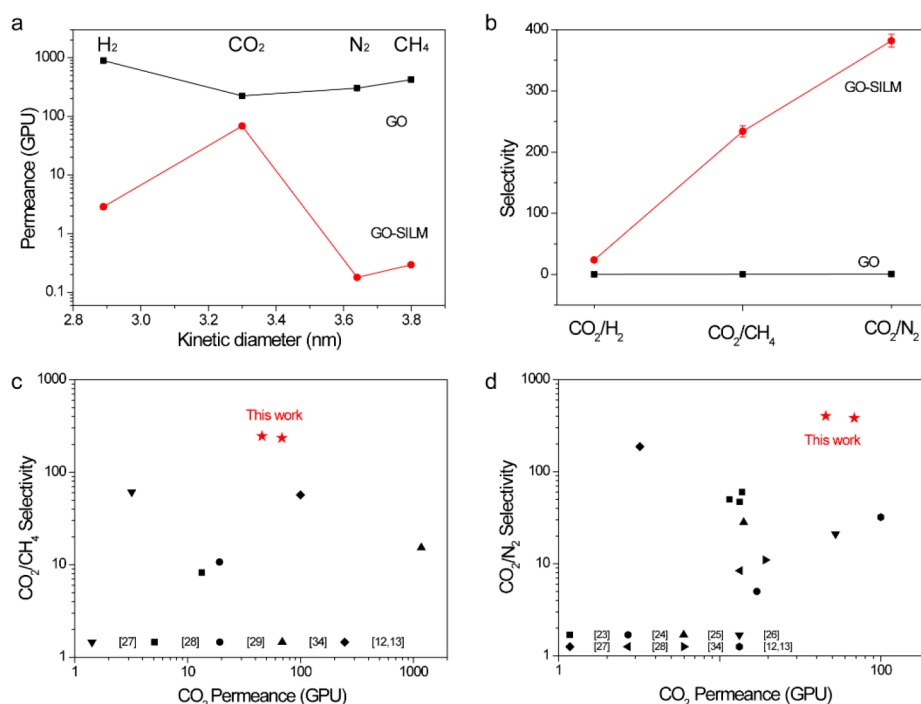


Figure 3. Gas separation performance of a GO-SILM with a thickness of 1050 nm. (a) Permeance and (b) selectivity of a GO membrane and GO-SILM. Error is within 10%. (c) Selectivities of CO_2/CH_4 and (d) CO_2/N_2 versus the permeance of CO_2 for state-of-the-art [BMIM][BF_4]-based and GO-based membranes, respectively. The corresponding data are listed in Table S5.

gas species.^{41–43} This is also applicable in our case for GO-SILM. The separation of CO_2/N_2 of GO-SILM, which is much higher than that of bulk [BMIM][BF_4], originated from the much enhanced solubility of CO_2 in GO-SILM due to the confinements of IL but less for N_2 . The solubility coefficient of CO_2 in GO-SILM based on time-lag method measurement is about 2 times that of bulk IL, as reported,^{41–43} which is significantly larger than that of other gases. This is also supported by the increment of binding energy between CO_2 to IL from bulk IL to GO-SILM by simulation and the significant increment of adsorption amount of CO_2 in GO-SILM compared to that in bulk ILs (Figure S11). The increment of the solubility of CO_2 should be due to the nanconfinement ILs in GO nanochannels. Therefore, in our case, the CO_2 /gas selectivity is mainly dominated by the solubility of gas species.

The gas transport pathway through a GO-SILM is schematically illustrated in Figure 4a. The center of mass distribution of [BMIM][BF_4] molecules confined in the GO nanochannels is plotted in Figure 4b,c, which demonstrates a layered structure of the nanoconfined IL at an interlayer d -spacing of 1.94 nm that aligns with the experimental measurements (see the details in the Calculation Section in Methods). According to the MD simulation results for the center-of-mass distribution, two minor [BMIM][BF_4] peaks are located at the center of the 1.94 nm channels, and one [BMIM][BF_4] layer is located near the GO walls. There is a shift between the peaks of cations and anions near the wall, due to the negatively charged nature of GO surfaces. The layered feature of cations and anions in the nanoconfined ILs agrees with recent studies on ILs confined in nanoporous carbon.³² Moreover, the center-of-mass distribution of CO_2 molecules in the GO-SILM overlaps with the minor peaks of the anions and cations at the center of the GO nanochannel (Figure 4c), where the density of IL is low and preferable for the CO_2 molecules to transport. On the other

hand, the CO_2 -IL interaction calculated from our MD simulations further shows that the binding energy between CO_2 and the [BMIM][BF_4] IL is the strongest among the four species under investigation (CO_2 , N_2 , H_2 , CH_4). Additionally, we find that the binding between CO_2 and the [BF_4][−] anion is stronger than that for CO_2 -cation binding, which is distinct from other gases due to the polarity of C=O bonds (Figure 4d and Figure S12). Therefore, the anion layer forms a fast lane for CO_2 transport, offering high selectivity and fast permeance as measured in our experiments, and these facts uncovered from MD simulations clearly demonstrate the solution-diffusion mechanism of gas transport and explain the contrast between permeance measured for CO_2 and other gas species.

The influence of the GO-SILM thickness on the separation performance was also investigated, as shown in Figure S13. GO-SILMs with different thicknesses were prepared by filtering different volumes of a GO/[BMIM][BF_4] dispersion (Figure S14). As the membrane thickness increases, the permeances of all the gases decrease, and the selectivities increase. Based on the trade-off between the permeance and the selectivity, the optimum thickness of the GO-SILM is 1050 nm.

Flue gas containing CO_2 is typically at relatively high temperatures (313.75–348.15 K).⁴⁴ Therefore, the thermal stability of a prepared GO-SILM with a thickness of 1050 nm was investigated by measuring the CO_2 permeance at different temperatures in the range from 293.15 to 373.15 K (Figure S15). The results show that the permeance of CO_2 in the GO-SILM increases with temperature (Table S9). The self-diffusion coefficient of CO_2 is reported to increase with temperature, whereas the CO_2 solubility decreases with temperature.^{22,23} In our case, the increase in the diffusion coefficient is more influential than the decrease in the solubility, which results in a net increase in the permeance of CO_2 (Table S9). Our MD simulation results show that the nanoconfinement reduces the

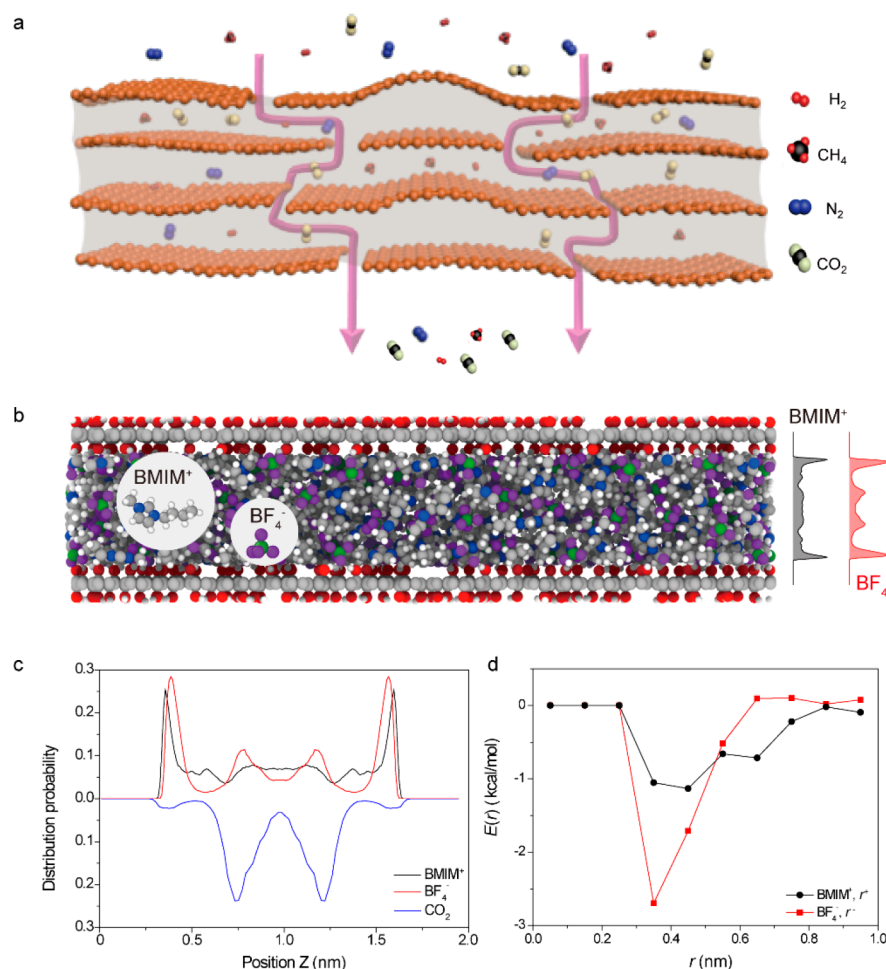


Figure 4. Gas–IL binding and gas transport pathways. (a) Schematic illustration of the gas solution–diffusion transport pathway through a GO-SILM; (b) molecular structures of [BMIM][BF₄] IL confined in the GO nanochannels. The green tetrahedrons represent the BF₄[−] anions. (c) Spatial distribution of the centers of masses of [BMIM]⁺, [BF₄][−], and CO₂ along the width direction of the GO nanochannel. (d) Interaction energy between gas (CO₂) and IL confined in the GO nanochannel, measured from simulation data as functions of the gas–cation ([BMIM]⁺) and gas–anion ([BF₄][−]) distances r^+ and r^- .

self-diffusion coefficients of all the gas species investigated, but the selectivity should not be significantly modified due to the nanoconfinement (Table S7). The selectivity of CO₂/N₂ at 373.15 K is 225, which is more than half that of the value at room temperature, and it is also competitive compared with those other supporting material.^{45–47} This performance is still above the Robeson upper bound. The permeance of CO₂ in a GO membrane decreases as the temperature increases due to the low sorption enthalpy, which agrees with previous results.¹² Flue gas also contains a certain water,⁴⁴ and to simulate the real condition, we measured the gas separation factor under different relative humidity (dry, 10, 30, and 60%) at 323 K. The results are shown in Table S10. The factor under 10% relative humidity was higher than the dry condition, as a low content of water in IL can increase the diffusion coefficient of CO₂. It might be because the water content increases, the solubility of CO₂ decreases, and the selectivity decreases.⁴⁸ To test the durability of the GO-SILM, the permeance of CO₂ was measured under 0.1 and 0.5 MPa for 9 days (Figure S16), and the selectivity of CO₂/N₂ is shown in Figure S17. The separation performance of the GO-SILM is nearly constant. The FTIR spectra of the GO-SILM after the gas separation and heating treatment show few changes (Figure S18), indicating

the long-term, ultrahigh thermal and mechanical stability of the membrane.

CONCLUSION

A high-performance CO₂-philic membrane, GO-SILM, was developed by nanoconfining [BMIM][BF₄] into the 2D nanochannels of a GO membrane. The nanoconfined [BMIM][BF₄] significantly facilitates the transportation of CO₂ and improves the selectivity for CO₂ over other gases. The high-temperature resistance, long-term durability, and high-pressure stability of GO-SILMs show the great potential of CO₂ separation and competitiveness to polymer-supported IL membranes. The facile filtration assembling process improves the potential of laminated membranes with nanoconfined ILs for gas separations.

METHODS

Synthesis of GO Nanosheets. The graphene oxide sheets were prepared by the modified Hummers method.⁴⁹ Graphite powder (0.5619 g) was added into a mixture of 9.9 mL of concentrated sulfuric acid, 1.875 g of phosphorus pentoxide, and 1.875 g of potassium persulfate under 80 °C in a water bath with stirring for 4.5 h. The mixture was diluted with 750 mL of deionized water and collected on a polycarbonate membrane with 200 nm pore size by vacuum filtration.

Afterward, it was washed with an excess of deionized water to remove the residual acid until the pH value of the filtrate approached neutrality. Then it was dried under room temperature for 24 h. After being dried, the preoxidized graphite was added into 90 mL of concentrated sulfuric acid in an ice–water bath with continuous stirring to mix evenly. While being fully dispersed, 11.25 g of potassium permanganate was gradually added. The mixture was transferred into a water bath under 35 °C with stirring for 2 h while the reaction did not release heat any more. After that, the mixture was diluted with 187.5 mL of deionized water in an ice–water bath. Then it was placed in a water bath at 35 °C for another 2 h. After that, 525 mL of deionized water and 15 mL of 30% hydrogen peroxide were added under stirring evenly. The color changed to brilliant yellow. After there was not any bubble produced, 30 mL of hydrochloric acid (37%) was added. Then the solution was placed in an ambient environment for 2 days. The sediment was collected and centrifuged with hydrochloric acid (10%) and deionized water by turns for five times followed by deionized water for five times to remove the residual acid and metal ions. The final product was dried under 50 °C in a blast air oven for 2 days.

Synthesis of GO Membrane and GO-SILM. By sonicating 0.2 g of the as-prepared GO in 1 L of deionized water for 1 h and stirring for 30 min, a stable brownish 0.2 mg/mL GO/water dispersion was obtained. The brownish 0.05 mg/mL GO/[BMIM][BF₄] dispersion was obtained by adding 2.5 mg of as-prepared GO in 50 mL of [BMIM][BF₄] and sonicating for 1 h and stirring for 30 min by turn for three times. The GO membrane and GO-SILM were fabricated by vacuum filtration on an AAO (Whatman) membrane with a pore size of 200 nm. The substrate of the resulting GO-SILM was washed by gently floating the AAO-supported GO-SILM on ethanol for 10 s, quickly dried by hot blowing to removing the effect of the ILs trapped in the channels of AAO, and then dried at 60 °C in a vacuum oven for 12 h. The C/O value tested by X-ray photoelectron spectroscopy shows a slight reduction of GO by increasing the C/O of 2.74 (before vacuum drying) to 3.01 after vacuum drying (Table S3). The gas performance of GO-SILM with substrate washed by ethanol and unwashed by ethanol is shown in Table S4.

Characterization. The scanning electron microscopy (SEM) images were obtained with a Hitachi S-4800 field-emission scanning electronic microscope equipped with an energy-dispersive X-ray spectrometer, and X-ray diffraction patterns were obtained with a Shimadzu XRD-6000 with a Cu K α radiation source. DSC results were obtained by TA Q200 differential scanning calorimeter, and ionic liquid contact was tested by an OCA 20 video-based contact angle measuring device. AFM images were obtained with a Bruker Dimension Edge instrument, and FTIR spectra were obtained with a Bruker Tensor 27. The C/O value was tested by a Kratos, AXIS Supra, X-ray photoelectron spectrometer. Gas adsorption isotherm was carried out at 298 K using a circulating thermostatic ethylene glycol bath; samples were not further treated prior to analysis. Isotherms up to 1 bar were measured on a Belsorp Max (MicroteacBEL Corporation, Japan).

All the NMR experiments were performed at a magnetic field strength of 9.4 T on a Bruker Avance III-HD 400 MHz NMR spectrometer at resonance frequencies of 400.13 MHz for ¹H and 100.61 MHz for ¹³C. The ¹³C cross-polarization (CP) contact time was 2 ms, and the ¹H decoupling power was 60 W (4.2 μ s); 20 480 transients were accumulated. The ¹³C high-power ¹H decoupling NMR spectra were measured with a 45° flip angle (3.95 μ s), 5 s recycle delay, and the ¹H decoupling power was 80 W (4.2 μ m). The ¹H–¹³C heteronuclear chemical shift correlation spectroscopy (HETCOR) 2D spectra were acquired by employing the conventional CP-HETCOR method with ¹³C detection. The radio frequency field values used for the CP transfers were identical to those employed for the acquisition of ¹³C CP–MAS spectra, and the ¹H homonuclear decoupling power was 80 W, with spectral widths of 20 and 50 kHz for ¹H and ¹³C dimensions, respectively. The radio frequency field values used for the CP transfers were identical to those employed for the acquisition of ¹³C CP–MAS spectra.

For the solid- or gel-state samples, 3.2 mm magic angle spinning probes were employed at a spinning speed of 15 kHz, whereas liquid samples were in the static state. The ¹³C signals were externally referenced to the methylene signal of adamantane at 38.5 ppm. All the reported NMR chemical shifts, along with the Results and Discussion and Supporting Information, have an estimated maximum interval of uncertainty equivalent to half the minimum visible parts per million scale interval (± 0.05 ppm).

Gas Separation. The gas permeance was measured by a bubble flow meter with a total volume of 1 mL with 0.01 mL accuracy. The GO-SILM was immobilized in an in-line stainless steel filter holder, with an effective membrane surface area of 2.2 cm², purchased from Millipore. The gas permeance measurements were performed more than 10 times for each membrane, and three membranes were tested with errors within $\pm 15\%$. Gas permeance (Q) was calculated by eq 1:¹²

$$Q = \frac{1}{P_u - P_d} \times \frac{273.15}{273.15 + T} \times \frac{P_{atm}}{101.325} \times \frac{1}{A} \times \frac{dV}{dt} \quad (1)$$

where P_u is the upstream pressure, P_d is the downstream pressure (atmosphere in our testing environment), P_{atm} is the atmospheric pressure, A is the membrane effective area, T is the temperature, and dV/dt is the volumetric displacement rate in the bubble flow meter. The 50% porosity of the AAO was considered. The ideal selective factor of two gases is the ratio of two gas permeances:¹²

$$\alpha = \frac{Q_1}{Q_2} \quad (2)$$

where Q_1 and Q_2 are the two permeance, respectively.

The gas permeability was calculated by multiplying permeance (Q) and thickness. The heating equipment was customized by the Shanghai Songdao Company.

The apparatus for CO₂, H₂, CH₄, and N₂ permeation test based on “time-lag” method was employed to calculate gas permeability (P), diffusivity (D), and solubility (S) by following eq 3 and 4,³⁹ as described in another study.⁵⁰

$$P = - \left(\frac{dp_1}{dt} \right)_{t \rightarrow \infty} \frac{V}{RT} \frac{l}{A} \frac{1}{(p_1 - p_2)} \quad (3)$$

$$D = \frac{l^2}{6\theta_L} \quad (4)$$

$$S = \frac{P}{D} \quad (5)$$

where p_1 is the downstream pressure, p_2 the upstream pressure, V the downstream volume, l the membrane thickness, A the available area for permeation, T the temperature, R the ideal gas constant, and θ_L the time lag.

An auxiliary device, containing two flow controllers, a pressure controller, and a humidometer, that imitates the device in other study⁵¹ was used to control humidity. The humidified mixed gas separation factor was measured by gas chromatography (FULI 9790). Before the test, the membrane was equilibrated at the desired relative humidity by filling the system with a certain pressure of water vapor to eliminate the influence of water in hydrous gas. The temperature was controlled under 323.15 K to simulate the real condition. Detailed data are shown in Table S10.

Calculation Section. The mass ratio of GO and [BMIM][BF₄] in the resulting GO-SILM membrane was calculated based on the thickness increment of the resulting membrane. As [BMIM][BF₄] is fully filled in the nanochannel of GO, the volume of interlayer space of GO and the increase of thickness are the volume of [BMIM][BF₄]. With V_1 set as the volume of interlayer space of GO, V_2 as the volume increment of resulted GO-SILM compared GO membrane, the mass of [BMIM][BF₄] is calculated by $(V_1 + V_2)\rho$, where ρ is the density of [BMIM][BF₄]. With the GO-SILM at a thickness of 1050 nm, containing 0.15 mg of GO, as an example,

$$V_1 = \frac{0.52}{0.87} \times 470 \times S \quad (6)$$

$$V_2 = (1050 - 470) \times S \quad (7)$$

where S is the membrane area, with diameter of 2 cm. As a result, the mass of [BMIM][BF₄] is 0.327 mg, and the mass ratio of GO and [BMIM][BF₄] in the resulting GO-SILM is 0.46.

The penetration of [BMIM][BF₄] obviously expanded the d -spacing of the GO membrane, with the constant layers N determined by h/d , where h and d are the thickness and interlayer d -spacing of the GO membrane at 470 and 0.87 nm, respectively. The interlayer d -spacing of GO-SILM is calculated by h'/N , where h' is the thickness of GO-SILM, 1050 nm, and the result is ~ 1.94 nm.

Capillary Force Estimation. The contact angle (θ_c) of [BMIM][BF₄] on the GO membrane is $\sim 30^\circ$ (Figure S3). The surface tension of [BMIM][BF₄] is measured to be 46.2 mN/m, whereas the lamellar space of GO nanosheets is 19.4 Å. The capillary force is estimated to be 82.5 MPa from the Young–Laplace equation. The capillary pressure of [BMIM][BF₄] in GO nanochannels (1.94 nm) is calculated to be 82.5 MPa from the Young–Laplace eq 8.³⁸

$$\Delta p = 2\gamma \cos \theta_c / \alpha \quad (8)$$

where Δp is the pressure difference across the fluid interface, γ is the surface tension, θ_c is the contact angle of the fluids and the solids in which they are contacted, and α is the radius of the circular pore.

Molecular Dynamics Simulation Details. The molecular structures of GO sheets contain oxygen-rich functional groups such as hydroxyl, epoxy, and carbonyl groups on the basal plane and edges of graphene. Compared to epoxy and carbonyl functionalization, it was reported that the hydroxyl groups can stay rich in the long-living quasi-equilibrium state of GO⁵² and thus are included in our model here. A typical fraction of hydroxyl-functionalized carbon atoms in the graphene lattice for GO, $c = n_{\text{OH}}/n_{\text{C}}$, is 20%,⁵³ whereas in our GO samples measured in experiments, the ratio is $\sim 10\%$. Here, n_{OH} and n_{C} are the numbers of hydroxyl groups and carbon atoms, respectively. We thus constructed graphene with hydroxyl groups on both sides, with $c = 10\%$. The carbon atoms functionalized by hydroxyl are sampled randomly. To simulate the nanochannels, we constructed a model consisting of two parallel GO sheets with periodic boundary conditions applied in the lateral directions only, with dimensions of $15.1 \text{ nm} \times 4.3 \text{ nm}$. The interlayer distance between GO sheets is fixed at $d = 1.94 \text{ nm}$, according to the measured value for our GO-SILMs.

Classical MD simulations were performed using the large-scale atomic/molecular massively parallel simulator (LAMMPS).⁵⁴ The all-atom optimized potentials for liquid simulations (OPLS-AA) are used for the GO structures,⁵⁵ which model essential many-body interatomic interactions.⁵⁶ The AMBER force field proposed by Cornell and co-workers are used for the [BMIM][BF₄] ionic liquids.⁵⁷ Several sets of parameters have been reported in the literature in functional forms of AMBER force field,^{58–61} and after benchmark tests, we employ the one reported by de Andrade and co-workers,⁵⁸ which could capture the structural and dynamical properties of [BMIM][BF₄]. The force field parameters for the gas molecules under investigation (CO₂, N₂, H₂, CH₄) are taken from our previous work.⁶² The intermolecular van der Waals interactions are modeled in the Lennard-Jones 12–6 form with heteroatomic pair parameters determined from the Lorentz–Berthelot mixing rule. The time step for integrating Newton's equations is 1 fs. The Ewald summation method⁶³ is used for the long-range electrostatic interactions, and the SHAKE algorithm is applied for all hydrogen-involved energy terms to avoid computation of the high-frequency vibrations that require shorter time steps. The interlayer gallery between GO sheets is filled with ILs by using the PACKMOL package.⁶⁴ The system is equilibrated by simulating annealing to 300 K, and the structure is then equilibrated for 3 ns, with the data in the last 2 ns used for analyzing thermodynamic information such as pressure. The Nosé–Hoover thermostat is used for temperature control.⁶³ By adjusting the number of IL molecules and the width of the channel, the anisotropic pressure of confined ILs can be minimized to $|p_z| < 50 \text{ MPa}$ and $|p_x|, |p_y| < 100 \text{ MPa}$ at $d = 1.94 \text{ nm}$.

After equilibration, another 5 ns run is simulated to produce data for analysis of the IL molecular structures. For the calculation of gas–IL binding energies and gas diffusivity, single gas molecules are added to the ILs, with further equilibration of 3 ns, and another data production run of 5 ns is carried out. Twenty subsets of simulation data are saved for every 100 ps in 10 independent runs for the thermodynamic averaging.

The gas–IL binding energies are calculated as

$$E_b = E_{\text{IL}} + E_{\text{gas}} - E_{\text{gas+IL}} \quad (9)$$

Here, E_{IL} is the energy of IL in nanochannels or the bulk form, E_{gas} is the energy of an isolated gas molecule, and $E_{\text{gas+IL}}$ is the energy of the complex with gas dissolved in the IL. Our simulation results show that the binding energies for all the gas species are positive and slightly higher comparable to the values measured in the bulk (Figure S10 and Table S7), and the binding of CO₂ to the IL is the strongest compared that with N₂, H₂, and CH₄. These results indicate the binding affinity of gases we modeled in nanoconfined ILs.

The self-diffusion coefficients of gas molecules are calculated from the Einstein relation, that is

$$D = \frac{1}{4} \lim_{t \rightarrow \infty} \frac{d}{dt} \langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle \quad (10)$$

Here, $\mathbf{r}(t)$ is the center of mass positions of the gas molecules. In our MD simulations of a few nanoseconds, the mean-square displacement (MSD) $\langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle$ is calculated on the time series, with average taken over data sets with different starting time in the series. The time-averaging of MSD is performed every 50 ps through recorded time series in the thermal equilibration state for gas molecules diffusing within the IL-filled GO nanochannel, with an almost linear relation between the MSD and t .⁶⁴

The absorption energy of a pair of IL molecules on the surfaces of graphene and GO channel, E_g , was calculated, which is -23.23 ± 0.94 and $-75.96 \pm 7.67 \text{ kcal/mol}$ per pair for the graphene and GO channels, respectively. The results suggest that the GO channel functionalized by oxygen-containing groups has significantly higher affinity to the IL, which is attributed the electrostatic interaction at the interfaces.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsnano.8b00367.

AFM, additional SEM images, XRD, FTIR results, photo images, comparison with other membranes, additional tables, additional simulation results, gas adsorption data, and additional gas separation performances (PDF)

AUTHOR INFORMATION

Corresponding Authors

*E-mail: kxq@zju.edu.cn.

*E-mail: xuzp@tsinghua.edu.cn.

*E-mail: pengxinsheng@zju.edu.cn.

ORCID

Ke Zhou: 0000-0003-2239-4381

Zhiping Xu: 0000-0002-2833-1966

Xinsheng Peng: 0000-0002-5355-4854

Author Contributions

W.Y., J.C., and K.Z. contributed equally to this work. W.Y. was responsible for the preparation and characterization of GO-SILMs membranes, evaluation of gas separation performance, and analysis. K.Z. and Z.X. were responsible for MD simulations. J.C. and X.K. contributed to the NMR measurement and analysis. D.C., Y.Y., and Y.G. contributed to the GO preparation and gas separation. W.Y., K.Z., J.C., and X.P. were

responsible for experimental design and manuscript preparation. X.P. was responsible for project planning. All the authors discussed the results and commented on the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (NSFC 21671171, 11472150, and 21573197), the National Basic Research Program of China 973 Program (2015CB655302), National Key Research and Development Program of China (2016YFA0200200 and 2016YFA0203600), and State Key Laboratory of Chemical Engineering No. SKL-ChE-16D03.

REFERENCES

- (1) Jacobson, M. Z. Review of Solutions to Global Warming, Air Pollution, and Energy Security. *Energy Environ. Sci.* **2009**, *2*, 148–173.
- (2) Brunetti, A.; Scura, F.; Barbieri, G.; Drioli, E. Membrane Technologies for CO₂ Separation. *J. Membr. Sci.* **2010**, *359*, 115–125.
- (3) Robeson, L. M. The Upper Bound Revisited. *J. Membr. Sci.* **2008**, *320*, 390–400.
- (4) Sumida, K.; Rogow, D. L.; Mason, J. A.; McDonald, T. M.; Bloch, E. D.; Herm, Z. R.; Bae, T. H.; Long, J. R. Carbon Dioxide Capture in Metal-Organic Frameworks. *Chem. Rev.* **2012**, *112*, 724–781.
- (5) Furukawa, H.; Cordova, K. E.; O’Keeffe, M.; Yaghi, O. M. The Chemistry and Applications of Metal-Organic Frameworks. *Science* **2013**, *341*, 1230444.
- (6) Shen, J.; Liu, G.; Huang, K.; Jin, W.; Lee, K. R.; Xu, N. Membranes with Fast and Selective Gas-transport Channels of Laminar Graphene Oxide for Efficient CO₂ Capture. *Angew. Chem.* **2015**, *127*, 588–592.
- (7) Liu, G.; Jin, W.; Xu, N. Graphene-based Membranes. *Chem. Soc. Rev.* **2015**, *44*, 5016–5030.
- (8) Tome, L. C.; Marrucho, I. M. Ionic liquid-based Materials: A Platform to Design Engineered CO₂ Separation Membranes. *Chem. Soc. Rev.* **2016**, *45*, 2785–2824.
- (9) Zhang, X.; Zhang, X.; Dong, H.; Zhao, Z.; Zhang, S.; Huang, Y. Carbon Capture with Ionic Liquids: Overview and Progress. *Energy Environ. Sci.* **2012**, *5*, 6668.
- (10) Cui, Y.; Kundalwal, S. I.; Kumar, S. Gas Barrier Performance of Graphene/Polymer Nanocomposites. *Carbon* **2016**, *98*, 313–333.
- (11) Li, H.; Song, Z.; Zhang, X.; Huang, Y.; Li, S.; Mao, Y.; Ploehn, H. J.; Bao, Y.; Yu, M. Ultrathin, Molecular-sieving Graphene Oxide Membranes for Selective Hydrogen Separation. *Science* **2013**, *342*, 95–98.
- (12) Kim, H. W.; Yoon, H. W.; Yoon, S. M.; Yoo, B. M.; Ahn, B. K.; Cho, Y. H.; Shin, H. J.; Yang, H.; Paik, U.; Kwon, S.; et al. Selective Gas Transport Through Few-layered Graphene and Graphene Oxide Membranes. *Science* **2013**, *342*, 91.
- (13) Kim, H. W.; Yoon, H. W.; Yoo, B. M.; Park, J. S.; Gleason, K. L.; Freeman, B. D.; Park, H. B. High-Performance CO₂-philic Graphene Oxide Membranes under Wet-Conditions. *Chem. Commun.* **2014**, *50*, 13563–13566.
- (14) Bates, E. D.; Mayton, R. D.; Ntai, I.; Davis, J. H. CO₂ Capture by a Task-Specific Ionic Liquid. *J. Am. Chem. Soc.* **2002**, *124*, 926–927.
- (15) Gupta, K. M.; Chen, Y.; Jiang, J. Ionic Liquid Membranes Supported by Hydrophobic and Hydrophilic Metal–Organic Frameworks for CO₂ Capture. *J. Phys. Chem. C* **2013**, *117*, 5792–5799.
- (16) Ban, Y.; Li, Z.; Li, Y.; Peng, Y.; Jin, H.; Jiao, W.; Guo, A.; Wang, P.; Yang, Q.; Zhong, C.; Yang, W. Confinement of Ionic Liquids in Nanocages: Tailoring The Molecular Sieving Properties of ZIF-8 for Membrane-Based CO₂ Capture. *Angew. Chem., Int. Ed.* **2015**, *54*, 15483–15487.
- (17) Yu, Y.; Mai, J.; Wang, L.; Li, X.; Jiang, Z.; Wang, F. Ship-In-A-Bottle Synthesis of Amine-functionalized Ionic Liquids in NaY Zeolite for CO₂ Capture. *Sci. Rep.* **2015**, *4*, 5997.
- (18) Yang, B.; Zhou, F.; Liu, S.; Wang, P.; Alshammari, A. S.; Deng, Y. Interaction between CO₂ and Ionic Liquids Confined in The Nanopores of SAPO-11. *RSC Adv.* **2015**, *5*, 48908–48915.
- (19) Li, X.; Zhang, L.; Zhou, D.; Liu, W.; Zhu, X.; Xu, Y.; Zheng, Y.; Zheng, C. Elemental Mercury Capture from Flue Gas by A Supported Ionic Liquid Phase Adsorbent. *Energy Fuels* **2017**, *31*, 714–723.
- (20) Peng, X.; Ichinose, I. Green-Chemical Synthesis of Ultrathin Beta-MnOOH Nanofibers for Separation Membranes. *Adv. Funct. Mater.* **2011**, *21*, 2080–2087.
- (21) Zhang, S.; Zhang, J.; Zhang, Y.; Deng, Y. Nanoconfined Ionic Liquids. *Chem. Rev.* **2017**, *117*, 6755–6833.
- (22) Jacquemin, J.; Costa Gomes, M. F.; Husson, P.; Majer, V. Solubility of Carbon Dioxide, Ethane, Methane, Oxygen, Nitrogen, Hydrogen, Argon, and Carbon Monoxide in 1-Butyl-3-Methylimidazolium Tetrafluoroborate between Temperatures 283 and 343 K and at Pressures Close to Atmospheric. *J. Chem. Thermodyn.* **2006**, *38*, 490–502.
- (23) Zhao, W.; He, G.; Zhang, L.; Ju, J.; Dou, H.; Nie, F.; Li, C.; Liu, H. Effect of Water in Ionic Liquid on The Separation Performance of Supported Ionic Liquid Membrane for CO₂/N₂. *J. Membr. Sci.* **2010**, *350*, 279–285.
- (24) Lee, J. H.; Hong, J.; Kim, J. H.; Kang, Y. S.; Kang, S. W. Facilitated CO₂ Transport Membranes Utilizing Positively Polarized Copper Nanoparticles. *Chem. Commun.* **2012**, *48*, 5298–300.
- (25) Ji, D.; Kang, Y. S.; Kang, S. W. Accelerated CO₂ Transport on Surface of AgO Nanoparticles in Ionic Liquid BMIMBF₄. *Sci. Rep.* **2015**, *5*, 16362.
- (26) Park, Y. S.; Ha, C.; Kang, S. W. Highly Permeable Ionic Liquid 1-Butyl-3-Methylimidazoliumtetrafluoroborate (BMIMBF₄)/CuO Composite Membrane for CO₂ Separation. *RSC Adv.* **2017**, *7*, 33568–33571.
- (27) Ghasemi Estahbanati, E.; Omidkhah, M.; Ebadi Amooghin, A. Interfacial Design of Ternary Mixed Matrix Membranes Containing Pebax 1657/Silver-Nanopowder/[BMIM][BF₄] for Improved CO₂ Separation Performance. *ACS Appl. Mater. Interfaces* **2017**, *9*, 10094–10105.
- (28) Choi, Y.; Hong, G. H.; Kang, S. W. Role of LiBF₄ in Ionic Liquid Membranes for Facilitated CO₂ Transport. *J. Nanosci. Nanotechnol.* **2016**, *16*, 2832–2835.
- (29) Hong, G. H.; Ji, D.; Kang, S. W. Facilitated CO₂ Transport and Barrier Effect Through Ionic Liquid Modified with Cyanuric Chloride. *RSC Adv.* **2014**, *4*, 16917.
- (30) Dai, Z.; Noble, R. D.; Gin, D. L.; Zhang, X.; Deng, L. Combination of Ionic Liquids with Membrane Technology: A New Approach for CO₂ Separation. *J. Membr. Sci.* **2016**, *497*, 1–20.
- (31) Hu, L.; Cheng, J.; Li, Y.; Liu, J.; Zhang, L.; Zhou, J.; Cen, K. Composites of Ionic Liquid and Amine-modified SAPO 34 Improve CO₂ Separation of CO₂-Selective Polymer Membranes. *Appl. Surf. Sci.* **2017**, *410*, 249–258.
- (32) Futamura, R.; Iiyama, T.; Takasaki, Y.; Gogotsi, Y.; Biggs, M. J.; Salanne, M.; Segalini, J.; Simon, P.; Kaneko, K. Partial Breaking of The Coulombic Ordering of Ionic Liquids Confined in Carbon Nanopores. *Nat. Mater.* **2017**, *16*, 1225.
- (33) Shiflett, M. B.; Yokozeki, A. Solubilities and Diffusivities of Carbon Dioxide in Ionic Liquids: [bmim][PF₆] and [bmim][BF₄]. *Ind. Eng. Chem. Res.* **2005**, *44*, 4453–4464.
- (34) Heo, J.; Choi, M.; Chang, J.; Ji, D.; Kang, S. W.; Hong, J. Highly Permeable Graphene Oxide/Polyelectrolytes Hybrid Thin Films for Enhanced CO₂/N₂ Separation Performance. *Sci. Rep.* **2017**, *7*, 456.
- (35) Marchetti, A.; Chen, J.; Pang, Z.; Li, S.; Ling, D.; Deng, F.; Kong, X. Understanding Surface and Interfacial Chemistry in Functional Nanomaterials via Solid-State NMR. *Adv. Mater.* **2017**, *29*, 1605895.
- (36) Katsyuba, S. A.; Zvereva, E. E.; Vidiš, A.; Dyson, P. J. Application of Density Functional Theory and Vibrational Spectroscopy toward The Rational Design of Ionic Liquids. *J. Phys. Chem. A* **2007**, *111*, 352–370.

- (37) Olivier-Bourbigou, H.; Magna, L.; Morvan, D. Ionic Liquids and Catalysis: Recent Progress from Knowledge to Applications. *Appl. Catal., A* **2010**, *373*, 1–56.
- (38) Peng, X.; Ichinose, I. Green-Chemical Synthesis of Ultrathin Beta-MnOOH Nanofibers for Separation Membranes. *Adv. Funct. Mater.* **2011**, *21*, 2080–2087.
- (39) Wu, X.; Tian, Z.; Wang, S.; Peng, D.; Yang, L.; Wu, Y.; Xin, Q.; Wu, H.; Jiang, Z. Mixed Matrix Membranes Comprising Polymers of Intrinsic Microporosity and Covalent Organic Framework for Gas Separation. *J. Membr. Sci.* **2017**, *528*, 273–283.
- (40) Knudsen, M. The Laws of The Molecular Current and The Internal Friction Current of Gases by Channels. *Ann. Phys.* **1909**, *333*, 75–130.
- (41) Blanchard, L. A.; Gu, Z.; Brennecke, J. F. High-Pressure Behavior of Liquid/CO₂ Systems. *J. Phys. Chem. B* **2001**, *105*, 2437–2444.
- (42) Camper, D.; Bara, J.; Koval, C.; Noble, R. Bulk-Fluid Solubility and Membrane Feasibility of Rmim-based Room-Temperature Ionic Liquid. *Ind. Eng. Chem. Res.* **2006**, *45*, 6279–6283.
- (43) Ramdin, M.; de Loos, T. W.; Vlucht, T. J. H. State-Of-The-Art of CO₂ Capture with Ionic Liquids. *Ind. Eng. Chem. Res.* **2012**, *51*, 8149–8177.
- (44) Kenarsari, S. D.; Yang, D.; Jiang, G.; Zhang, S.; Wang, J.; Russell, A. G.; Wei, Q.; Fan, M. Review of Recent Advances in Carbon Dioxide Separation and Capture. *RSC Adv.* **2013**, *3*, 22739.
- (45) Liu, Y.-F.; Xu, Q.-Q.; Cai, P.; Zhen, M.-Y.; Wang, X.-Y.; Yin, J.-Z. Effects of Operating Parameters and Ionic Liquid Properties on Fabrication of Supported Ionic Liquid Membranes Based on Mesoporous γ -Al₂O₃ Supports. *J. Membr. Sci.* **2018**, *545*, 176–184.
- (46) Abdelrahim, M. Y. M.; Martins, C. F.; Neves, L. A.; Capasso, C.; Supuran, C. T.; Coelho, I. M.; Crespo, J. G.; Barboiu, M. Supported Ionic Liquid Membranes Immobilized with Carbonic Anhydrases for CO₂ Transport at High Temperatures. *J. Membr. Sci.* **2017**, *528*, 225–230.
- (47) Santos, E.; Albo, J.; Irabien, A. Acetate Based Supported Ionic Liquid Membranes (SILMs) for CO₂ Separation: Influence of The Temperature. *J. Membr. Sci.* **2014**, *452*, 277–283.
- (48) Zhao, W.; He, G.; Zhang, L.; Ju, J.; Dou, H.; Nie, F.; Li, C.; Liu, H. Effect of Water in Ionic Liquid on The Separation Performance of Supported Ionic Liquid Membrane for CO₂/N₂. *J. Membr. Sci.* **2010**, *350*, 279–285.
- (49) Huang, H.; Mao, Y.; Ying, Y.; Liu, Y.; Sun, L.; Peng, X. Salt Concentration, pH and Pressure Controlled Separation of Small Molecules Through Lamellar Graphene Oxide Membranes. *Chem. Commun.* **2013**, *49*, 5963–5965.
- (50) Minelli, M.; Baschetti, M. G.; Doghieri, F.; Ankerfors, M.; Lindström, T.; Siró, I.; Plackett, D. Investigation of Mass Transport Properties of Microfibrillated Cellulose (MFC) Films. *J. Membr. Sci.* **2010**, *358*, 67–75.
- (51) Venturi, D.; Grupkovic, D.; Sisti, L.; Baschetti, M. G. Effect of Humidity and Nanocellulose Content on Polyvinylamine-Nanocellulose Hybrid Membranes for CO₂ Capture. *J. Membr. Sci.* **2018**, *548*, 263–274.
- (52) Kim, S.; Zhou, S.; Hu, Y.; Acik, M.; Chabal, Y. J.; Berger, C.; De Heer, W. A.; Bongiorno, A.; Riedo, E. Room Temperature Metastability of Multilayer Graphene Oxide Films. *Nat. Mater.* **2012**, *11*, 544.
- (53) Dreyer, D. R.; Park, S.; Bielawski, C. W.; Ruoff, R. S. The Chemistry of Graphene Oxide. *Chem. Soc. Rev.* **2010**, *39*, 228–240.
- (54) Plimpton, S. Fast Parallel Algorithms for Short-Range Molecular-Dynamics. *J. Comput. Phys.* **1995**, *117*, 1–19.
- (55) Jorgensen, W. L.; Maxwell, D. S.; TiradoRives, J. Development and Testing of The OPLS All-Atom Force Field on Conformational Energetics and Properties of Organic Liquids. *J. Am. Chem. Soc.* **1996**, *118*, 11225–11236.
- (56) Shih, C. J.; Lin, S.; Sharma, R.; Strano, M. S.; Blankschtein, D. Understanding the pH-Dependent Behavior of Graphene Oxide Aqueous Solutions: A Comparative Experimental and Molecular Dynamics Simulation Study. *Langmuir* **2012**, *28*, 235–241.
- (57) Cornell, W. D.; Cieplak, P.; Bayly, C. I.; Gould, I. R.; Merz, K. M.; Ferguson, D. M.; Spellmeyer, D. C.; Fox, T.; Caldwell, J. W.; Kollman, P. A. A Second Generation Force Field for The Simulation of Proteins, Nucleic Acids, and Organic Molecules. *J. Am. Chem. Soc.* **1995**, *117*, 5179–5197.
- (58) de Andrade, J.; Boes, E. S.; Stassen, H. Computational Study of Room Temperature Molten Salts Composed by 1-Alkyl-3-Methylimidazolium Cations-Force-Field Proposal and Validation. *J. Phys. Chem. B* **2002**, *106*, 13344–13351.
- (59) Wu, X.; Liu, Z.; Huang, S.; Wang, W. Molecular Dynamics Simulation of Room-Temperature Ionic Liquid Mixture of [bmim]-[BF₄] and Acetonitrile by A Refined Force Field. *Phys. Chem. Chem. Phys.* **2005**, *7*, 2771–2779.
- (60) Tsuzuki, S.; Shinoda, W.; Saito, H.; Mikami, M.; Tokuda, H.; Watanabe, M. Molecular Dynamics Simulations of Ionic Liquids: Cation and Anion Dependence of Self-Diffusion Coefficients of Ions. *J. Phys. Chem. B* **2009**, *113*, 10641–10649.
- (61) Liu, Z. P.; Huang, S. P.; Wang, W. C. A Refined Force Field for Molecular Simulation of Imidazolium-based Ionic Liquids. *J. Phys. Chem. B* **2004**, *108*, 12978–12989.
- (62) Jiao, S.; Xu, Z. Selective Gas Diffusion in Graphene Oxides Membranes: A Molecular Dynamics Simulations Study. *ACS Appl. Mater. Interfaces* **2015**, *7*, 9052–9059.
- (63) Allen, M. P.; Tildesley, D. J. *Computer Simulations of Liquids*; Clarendon Press: Oxford, 1987.
- (64) Jiao, S.; Xu, Z. Non-Continuum Intercalated Water Diffusion Explains Fast Permeation Through Graphene Oxide Membranes. *ACS Nano* **2017**, *11*, 11152–11161.